

Adaptive Hybrid Retrieval-Augmented Generation

Optimizing the computational cost-performance trade-off via learned query routing

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Graph-enhanced retrieval answers harder questions, but every query pays for it

Naive vector search only

10.9 s

mean time per query

≈3,500

model tokens per query*

2.2×
time

5.2×
tokens

Mix vector + knowledge graph

24.4 s

mean time per query

≈17,900

model tokens per query*

Yet Mix was needed on only **327 of 2,000** judged questions (16%). On 959 (48%), Naive was already enough.

Times: all 193 test questions. *Tokens: per-mode means over 15 instrumented queries (prompt, output and reasoning tokens).

Some questions need the graph, most do not

Q Which film came out first, The Love Route or Engal Aasan?

Gold answer: **The Love Route**

correct **Naive:** *"…The Love Route came out significantly earlier than Engal Aasan." (1915 vs 2009)*

correct **Mix:** *"…The Love Route was released significantly earlier than Engal Aasan."*

Label: naive_enough → route to Naive

Q When was the director of film Private Hell 36 born?

Gold answer: **October 26, 1912**

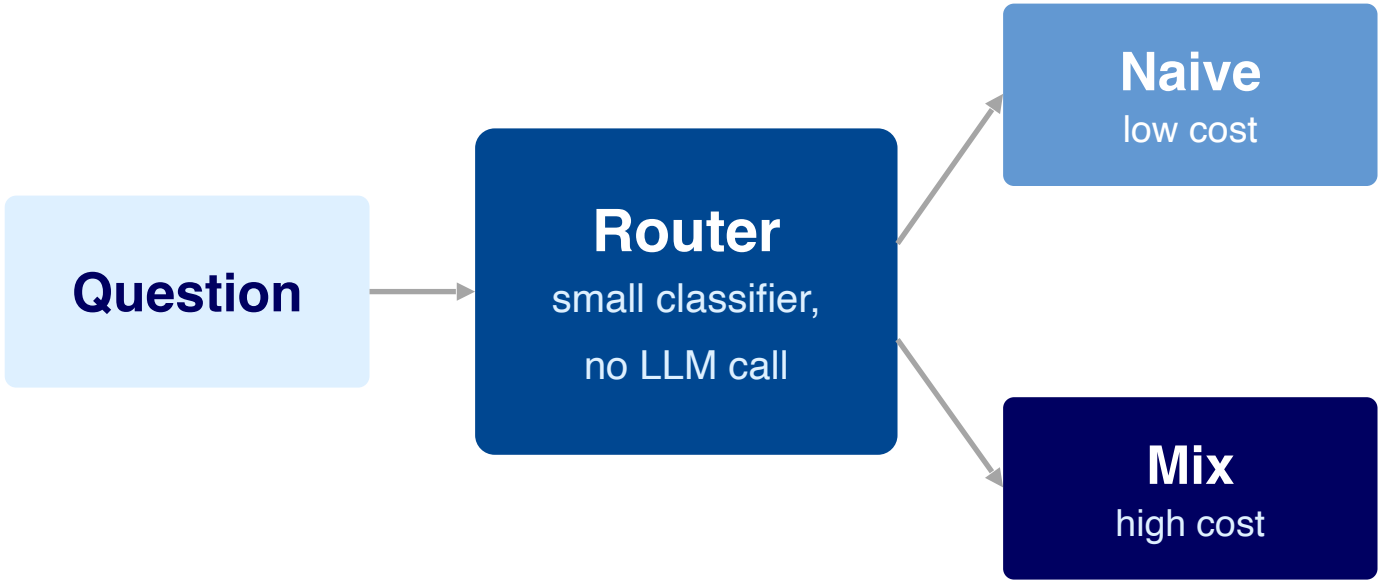
not enough **Naive:** *"…was directed by Don Siegel. However, the provided documents do not contain information regarding Don Siegel's birth date."*

correct **Mix:** *"…Donald Siegel was born on October 26, 1912."*

Label: mix_required → route to Mix

Real questions and answers from the 2WikiMultihopQA runs (abridged). Comparison question left, two-hop compositional question right.

The idea: let a cheap classifier pick the mode from the question, before retrieval runs



Trained on labels that say which mode was enough for each question (next section).

–42%
mean time per query vs. always running Mix

–63%
estimated model tokens per query

≈92%
of always-Mix's judged-correct answers kept*

ModernBERT router on 193 test questions. *Assumes Mix is correct wherever Naive was.

Can a lightweight classifier pick the cheapest sufficient retrieval mode?

Within one fixed Hybrid RAG setting (LightRAG, Naive vs. Mix), how well can a query-only classifier predict the minimum sufficient retrieval mode, and what does routing with it save?

Sub-question 1

Prediction

How does a fine-tuned ModernBERT classifier compare to a TF-IDF baseline and a question-type heuristic in predicting the label?

Sub-question 2

Cost

How much mean time and model-token cost does routing save compared with always-Mix, and what does it risk in answer quality?

Prior work routes between models or retrieval depth; this thesis routes between vector and graph retrieval

Adaptive-RAG Jeong et al., 2024	A question-complexity classifier picks no, single-step or multi-step retrieval.
RouterBench Hu et al., 2024	Routes each query between language models on cost vs. quality.
GraphRAG, LightRAG Edge et al., 2025; Guo et al., 2025	Graph-enhanced retrieval improves multi-hop answers at higher cost.

This thesis A query-only router choosing between two retrieval modes inside one LightRAG index, trained on judge-derived labels and evaluated with paired statistical tests.

Method

From benchmark questions to labels and routers

Both modes run on every question, and an LLM judge records which one was enough

LABELLING (offline, once)

2,000 questions
2WikiMultihopQA, multi-hop by design

→

LightRAG index
shared corpus: vector chunks + entity-relation graph

→

Run both modes
Naive and Mix answers, time per query

→

LLM-as-a-Judge
compares both answers with the gold answer

LEARNING AND EVALUATION

Binary dataset
1,286 questions: 900 train / 193 val / 193 test

→

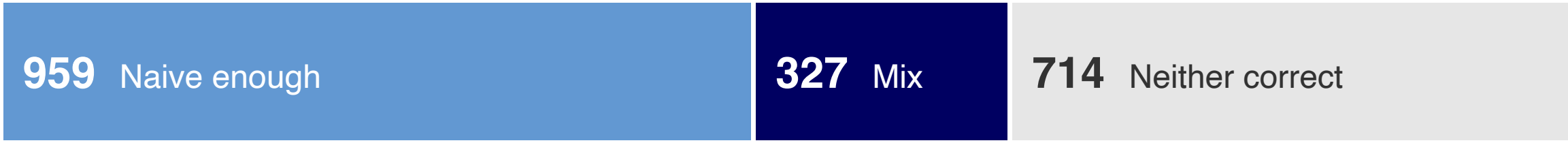
Train routers
TF-IDF + logistic regression; ModernBERT (fine-tuned)

→

Evaluate
as classifiers and as routing policies on cost

Same Gemini model generates and judges, temperature 0, 5 text chunks per mode; first 2,000 queryable questions of the 2WikiMultihopQA train split, in benchmark order.

The label is the cheapest mode that was judged sufficient



2,000 judged questions

Naive answer correct	**naive_enough**	label 0
Naive not enough, Mix correct	**mix_required**	label 1
Neither correct: routing cannot help	**none_enough**	excluded

1,286 questions

25% Mix required

split 900 / 193 / 193, stratified by question type and label

If both modes are correct, the judge is told to choose Naive: the label records the minimum sufficient mode, not the better-looking answer.

Two query-only routers: a lexical baseline and a fine-tuned encoder

TF-IDF + logistic regression

- Word and word-pair counts, weighted by rarity
- Balanced class weights
- Trains in seconds on a CPU
- Tests how much of the signal is lexical

Fine-tuned ModernBERT

- Pretrained encoder + classification head
- Class-balanced loss, early stopping
- 3 seeds averaged into one ensemble
- Trains in hours on a CPU

Both: threshold chosen on validation F1, then frozen. **Reference:** type-only rule (Mix if the question is compositional); not deployable, since real queries have no type label.

Each router is evaluated as a classifier and as a routing policy, with paired tests

Classification

- AUPRC on mix_required (primary, chance 0.254)
- Precision, recall, F1
- Routing-label accuracy

Routing policy

- Mean routed time (measured)
- Model tokens (estimated per mode)
- Under-routing: missed Mix call (answer risk)
- Over-routing: unneeded Mix call (wasted cost)

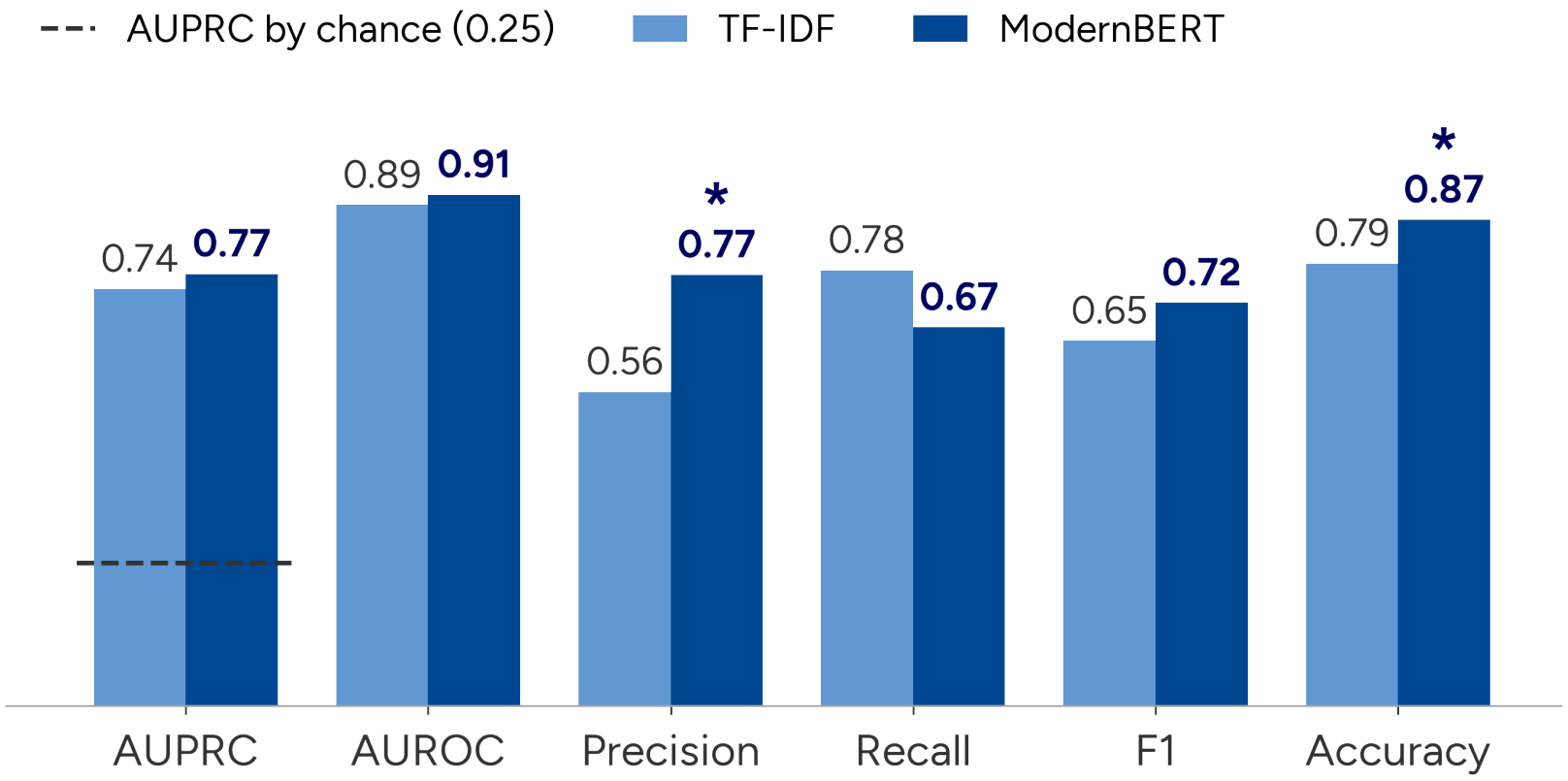
Statistics

- Same 193 test questions for all routers, so tests are paired
- Paired bootstrap 95% intervals
- Permutation tests, McNemar
- Holm correction (7 metrics)

Results

Prediction (sub-question 1), then cost and risk (sub-question 2), then where the signal comes from

Both routers find Mix-needing questions far above chance, and ModernBERT is more precise

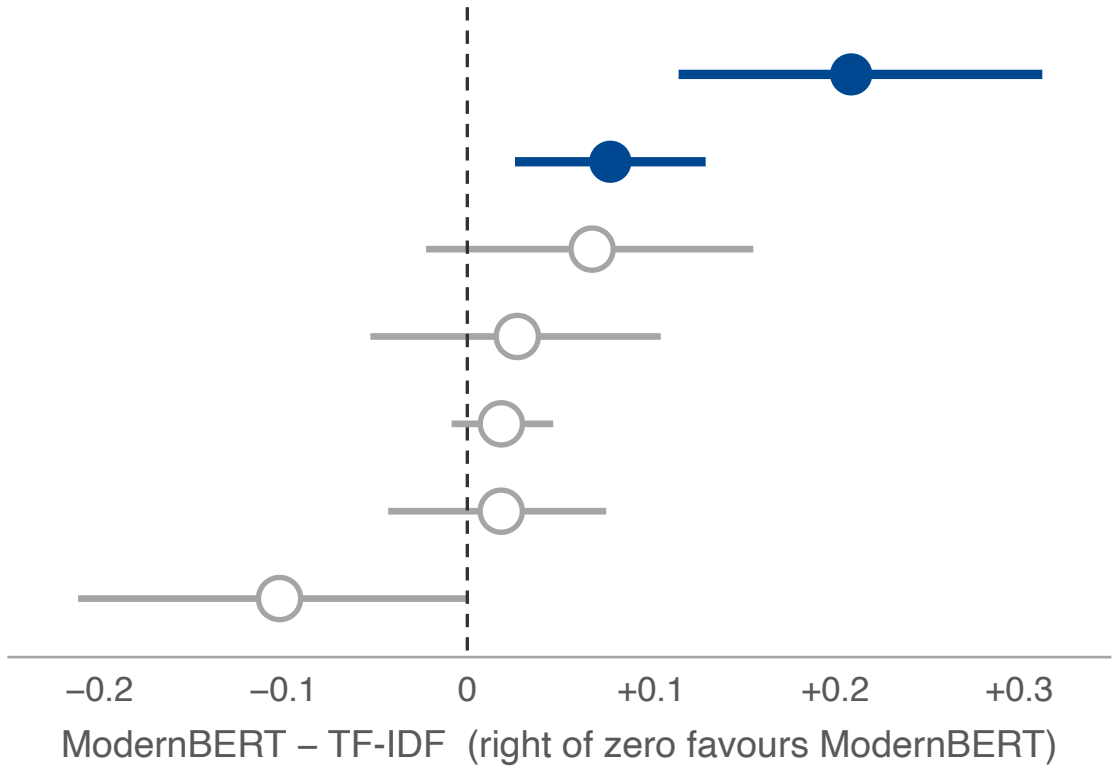


Chance level
AUPRC 0.254
AUROC 0.5

Significant (Holm) *
Precision, accuracy

Recall is lower
0.67 vs 0.78, not significant

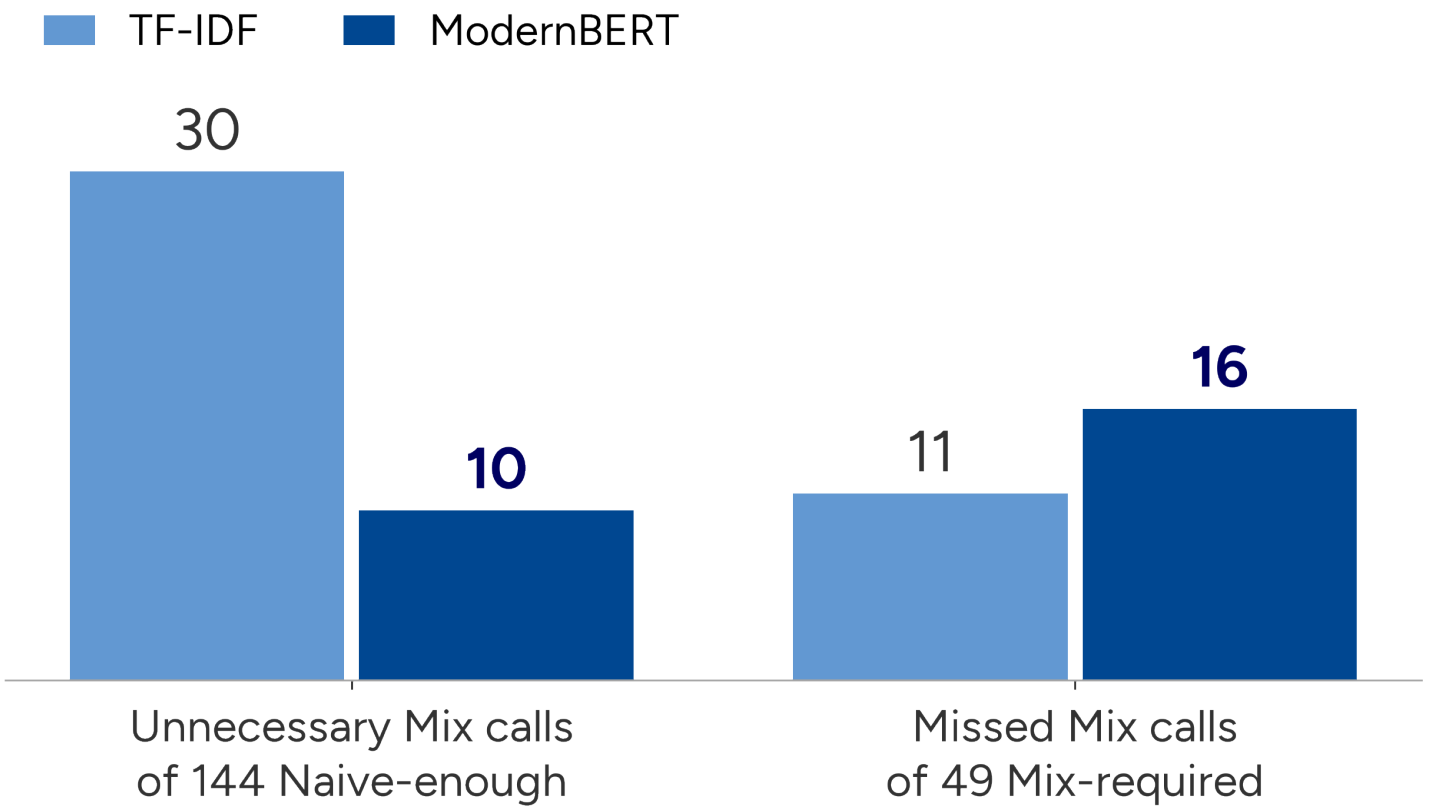
Paired tests: the precision and accuracy gains are significant; the ranking gains are not resolved



	difference	[95% interval]	Holm p
Precision	**+0.209**	**[+0.115, +0.312]**	**<0.001**
Accuracy	**+0.078**	**[+0.026, +0.130]**	**0.035**
F1	+0.068	[–0.022, +0.155]	0.56
AUPRC	+0.027	[–0.053, +0.105]	1.00
AUROC	+0.018	[–0.009, +0.047]	0.56
Balanced accuracy	+0.018	[–0.043, +0.075]	1.00
Recall	–0.102	[–0.212, +0.000]	0.56

Reading: ModernBERT is the more selective router. Whether it also ranks questions better cannot be decided with 49 Mix-required test questions.

ModernBERT makes a third as many unnecessary Mix calls, at a small cost in missed ones



Over-routing: wasted cost

30 → 10 unnecessary Mix calls

McNemar: 20 vs 0, $p < 0.001$

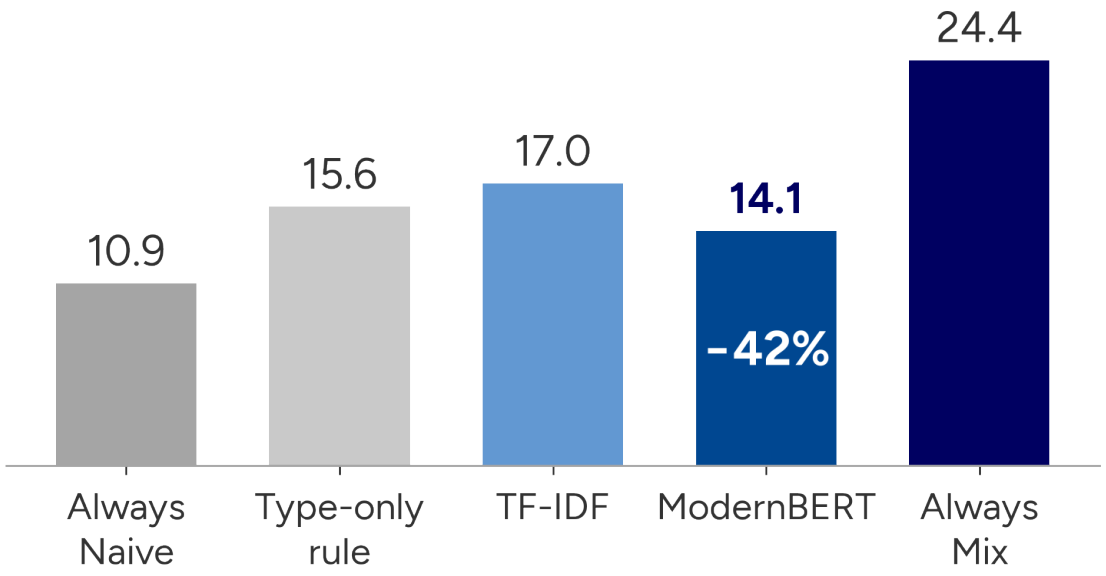
Under-routing: answer risk

11 → 16 missed Mix calls

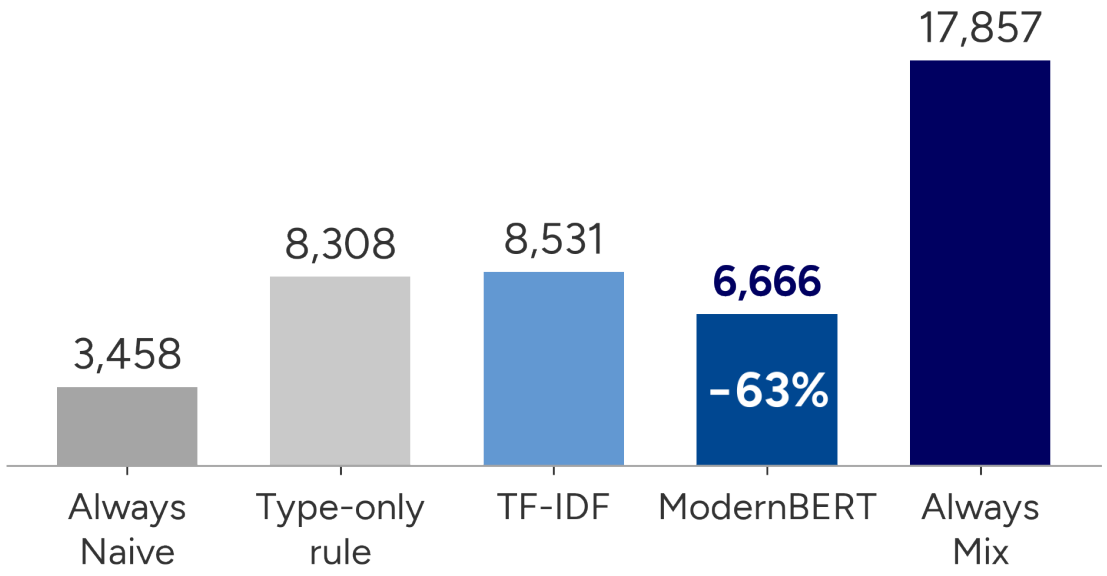
McNemar: 6 vs 1, $p = 0.125$ (not significant)

ModernBERT routing cuts mean time by 42% and estimated tokens by 63% compared with always-Mix

Mean time per query, seconds (measured)



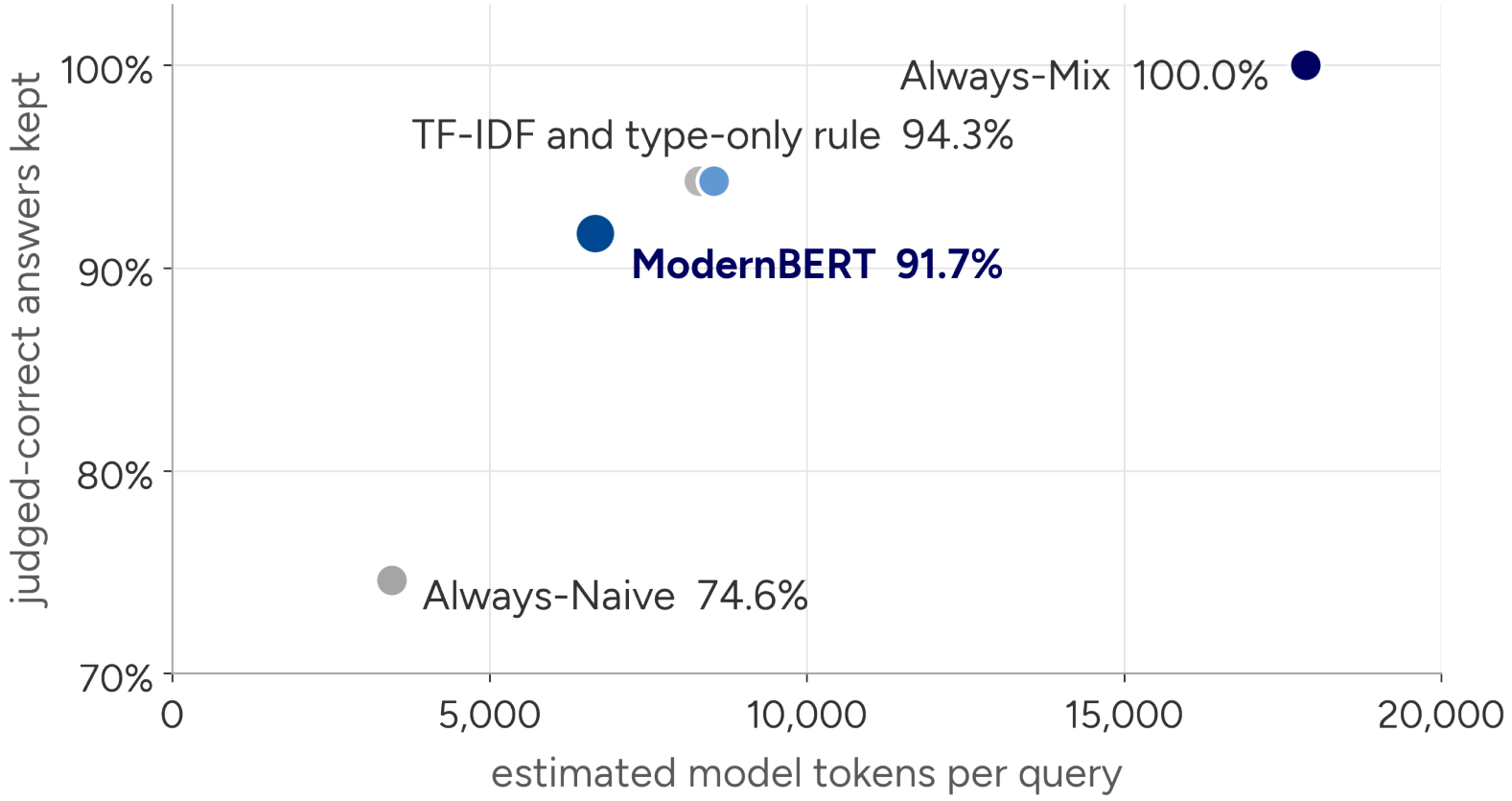
Model tokens per query (estimated*)



ModernBERT vs TF-IDF time: –2.9 s per query, 95% interval [–5.6, –0.9], p = 0.003.

*Tokens: per-mode means from 15 instrumented queries, no interval. The router's own inference time is not included in either chart.

ModernBERT keeps about 92% of judged-correct answers at 37% of always-Mix's tokens



ModernBERT

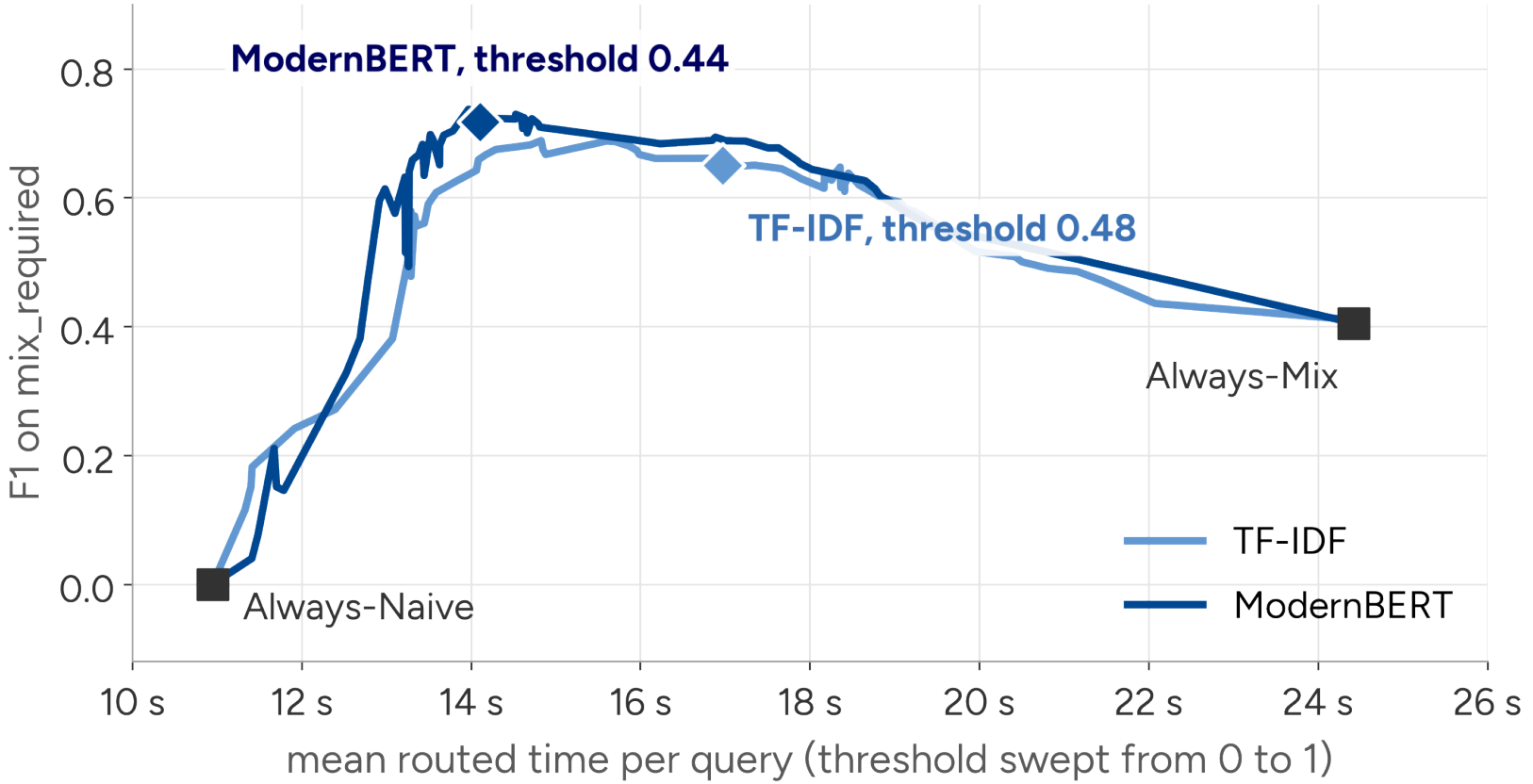
keeps 91.7% of the answers at 37% of always-Mix's token cost

TF-IDF

keeps 94.3% at 48%

Assumes Mix is correct wherever Naive was, so kept answers = 1 – under-routing. Answers were not re-judged per policy.

The threshold is a cost dial, and both chosen operating points sit well above the static policies



Reading the curve

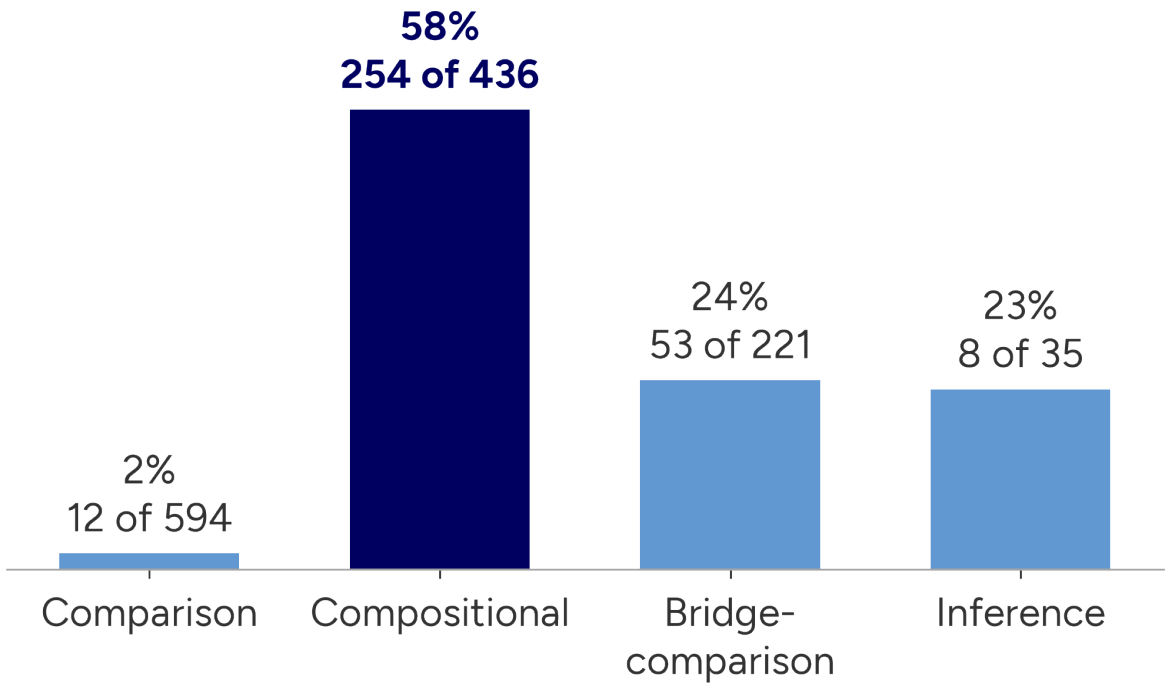
Each point is one threshold; lowering it sends more questions to Mix.

ModernBERT's chosen threshold sits near its peak; TF-IDF's is a few hundredths below its own (0.65 vs 0.69).

In deployment the threshold sets the cost.

Much of the signal is question structure: compositional questions are the ones that usually need Mix

Mix-required rate by question type (all 1,286 labelled questions)



Type-only rule

F1 0.67, using the benchmark's type label, which real queries do not have

ModernBERT vs the rule (text only)

Precision 0.77 vs 0.58 (Holm p = 0.002); accuracy higher only before correction

Within compositional questions

ModernBERT precision 0.78 vs TF-IDF 0.64 (p = 0.003, exploratory)

Weak spot: bridge-comparison

ModernBERT finds 1 of 8 Mix-required questions (backup 26)

What this means for a deployment like Ericsson's

Cheap to run, tunable

No language-model call at query time; the threshold sets the cost. Trains in hours on a CPU. Labels cost one run of both modes plus the judge per training question.

Validate on the target workload

Part of the signal is benchmark structure. Real queries need their own labels: a small human-labelled set or a judge that does not rely on gold answers.

Routing cannot fix everything

36% of the questions failed in both modes. They need a better answering pipeline or a fallback, not a better router.

Limitations, and what would remove them

Limitation	Next step
Labels come from an LLM judge, using the same Gemini model that wrote the answers; no human check	Hand-validate a sample; re-judge with a second model family
Token cost estimated from 15 instrumented queries	Log tokens for every query
One benchmark (chosen partly for label balance), one framework, row-random split	Template-grouped split; HotpotQA, MuSiQue; other frameworks
193 test questions (49 Mix-required) cannot resolve small ranking differences	A test set 5 to 20 times larger
Kept answers assume Mix is correct wherever Naive was	Re-judge the answer each policy picks

Conclusions

- **Prediction (sub-question 1).** From the question alone, both routers find Mix-needing questions at about 3× chance. ModernBERT is significantly more precise than TF-IDF (0.77 vs 0.56) and than the type-only rule; ranking gains are unresolved.
- **Cost (sub-question 2).** ModernBERT routing cuts mean time by 42% and estimated tokens by 63% versus always-Mix, keeping about 92% of judged-correct answers (assuming Mix is correct wherever Naive was).
- **Scope.** One benchmark, one framework, LLM-judged labels: validate on the target workload before deployment.

Contributions

- A study of query-only routing between a vector and a graph-enhanced retrieval mode within one RAG framework
- A judge-labelled routing dataset and an open, reproducible pipeline
- Evidence, from paired tests, that the encoder's gain lies in precision and therefore in cost

Thank you. Questions are welcome.

github.com/Dantecool02/Thesis_ericsson

Backup slides

Type the slide number in Keynote and press Return to jump

24 Full statistics table
25 Seed stability
26 Per question type
27 Cost and errors per policy
28 Dataset composition
29 Judge prompt
30 PR and ROC curves
31 Statistical power
32 Query-only vs Naive-first cascade
33 Configuration
34 Use of generative AI

Backup: full classification comparison with intervals and paired tests

Metric	TF-IDF [95% CI]	ModernBERT [95% CI]	Difference [95% paired CI]	p	Holm p
AUPRC	0.742 [0.610, 0.853]	0.769 [0.645, 0.889]	+0.027 [–0.053, +0.105]	0.518	1.000
AUROC	0.892 [0.836, 0.936]	0.910 [0.865, 0.949]	+0.018 [–0.009, +0.047]	0.165	0.563
Balanced accuracy	0.784 [0.718, 0.848]	0.802 [0.730, 0.872]	+0.018 [–0.043, +0.075]	0.520	1.000
Precision (mix)	**0.559 [0.437, 0.681]**	**0.767 [0.644, 0.889]**	**+0.209 [+0.115, +0.312]**	**<0.001**	**<0.001**
Recall (mix)	0.776 [0.650, 0.885]	0.673 [0.540, 0.809]	–0.102 [–0.212, +0.000]	0.127	0.563
F1 (mix)	0.650 [0.547, 0.740]	0.717 [0.607, 0.810]	+0.068 [–0.022, +0.155]	0.113	0.563
Accuracy	**0.788 [0.731, 0.845]**	**0.865 [0.819, 0.912]**	**+0.078 [+0.026, +0.130]**	**0.006**	**0.035**

n = 193 test questions (49 mix_required). Marginal CIs: 1,000 bootstrap resamples. Paired CIs: 10,000 paired resamples. p: paired permutation test (10,000). Holm over the 7 metrics. DeLong AUROC p = 0.18.

Backup: every ModernBERT seed scores above TF-IDF on the aggregate metrics

Run	Threshold	AUPRC	AUROC	Bal. acc.	Precision	Recall	F1	Accuracy
Seed 7	0.36	0.754	0.895	0.791	0.648	0.714	0.680	0.829
Seed 13	0.42	0.783	0.915	0.792	0.762	0.653	0.703	0.860
Seed 42	0.24	0.771	0.915	0.812	0.692	0.735	0.713	0.850
Sample std	–	0.015	0.011	0.012	0.057	0.042	0.017	0.016
Ensemble	**0.44**	**0.769**	**0.910**	**0.802**	**0.767**	**0.673**	**0.717**	**0.865**
TF-IDF	0.48	0.742	0.892	0.784	0.559	0.776	0.650	0.788

Point estimates. Aggregate metrics vary little across seeds; precision and recall vary more because each seed selects its own validation threshold. The ensemble costs three encoder forward passes per query.

Backup: per-question-type results on the test set

Slice	n	n mix	Router	AUPRC	Precision	Recall	F1	Accuracy
Comparison	90	2	TF-IDF	0.277	0.000	0.000	0.000	0.978
			ModernBERT	0.140	0.000	0.000	0.000	0.978
Compositional	65	38	TF-IDF	0.814	0.636	0.921	0.753	0.646
			ModernBERT	0.834	0.780	0.842	0.810	0.769
Bridge-comparison	33	8	TF-IDF	0.418	0.273	0.375	0.316	0.606
			ModernBERT	0.396	0.500	0.125	0.200	0.758
Inference	5	1	TF-IDF	0.333	0.000	0.000	0.000	0.400
			ModernBERT	0.500	0.000	0.000	0.000	0.800

Exploratory, unadjusted tests on compositional and bridge-comparison only: just compositional precision differs significantly ($p = 0.003$). Comparison and inference have too few Mix-required questions to test.

Backup: routed cost and routing errors per policy

Policy	To Mix	Mean time (s)	Est. tokens	Token saving	Missed Mix	Extra Mix	Kept answers*
Always-Naive	0.0%	10.95	3,458	80.6%	49 (0.254)	0 (0.000)	74.6%
Type-only rule	33.7%	15.59	8,308	53.5%	11 (0.057)	27 (0.140)	94.3%
TF-IDF	35.2%	16.98	8,531	52.2%	11 (0.057)	30 (0.155)	94.3%
ModernBERT	**22.3%**	**14.11**	**6,666**	**62.7%**	**16 (0.083)**	**10 (0.052)**	**91.7%**
Always-Mix	100.0%	24.42	17,857	0.0%	0 (0.000)	144 (0.746)	100.0%

Time: measured per query. Tokens: per-mode means from 15 instrumented queries (3,458 Naive, 17,857 Mix; prompt, output and reasoning tokens; embeddings excluded). Missed Mix = under-routing, Extra Mix = over-routing. *Kept answers = 1 – under-routing, assuming Mix is correct wherever Naive was.

Backup: dataset composition by split and question type

Split	Total	Comparison	Compositional	Bridge-comparison	Inference
Train	900	415 (8 mix)	305 (178 mix)	155 (37 mix)	25 (6 mix)
Validation	193	89 (2 mix)	66 (38 mix)	33 (8 mix)	5 (1 mix)
Test	193	90 (2 mix)	65 (38 mix)	33 (8 mix)	5 (1 mix)
Overall	**1,286**	**594 (12 mix)**	**436 (254 mix)**	**221 (53 mix)**	**35 (8 mix)**

Label flow: 2,000 judged → 959 naive_enough, 327 mix_required, 714 none_enough (excluded) → 1,286 binary.

Query set: first 2,000 of the 2,279 queryable questions of the 2WikiMultihopQA train split, in benchmark order. Split jointly stratified by question type and label.

Backup: the judge sees the gold answer and must return the minimum sufficient label

Labels defined in the prompt (quoted)

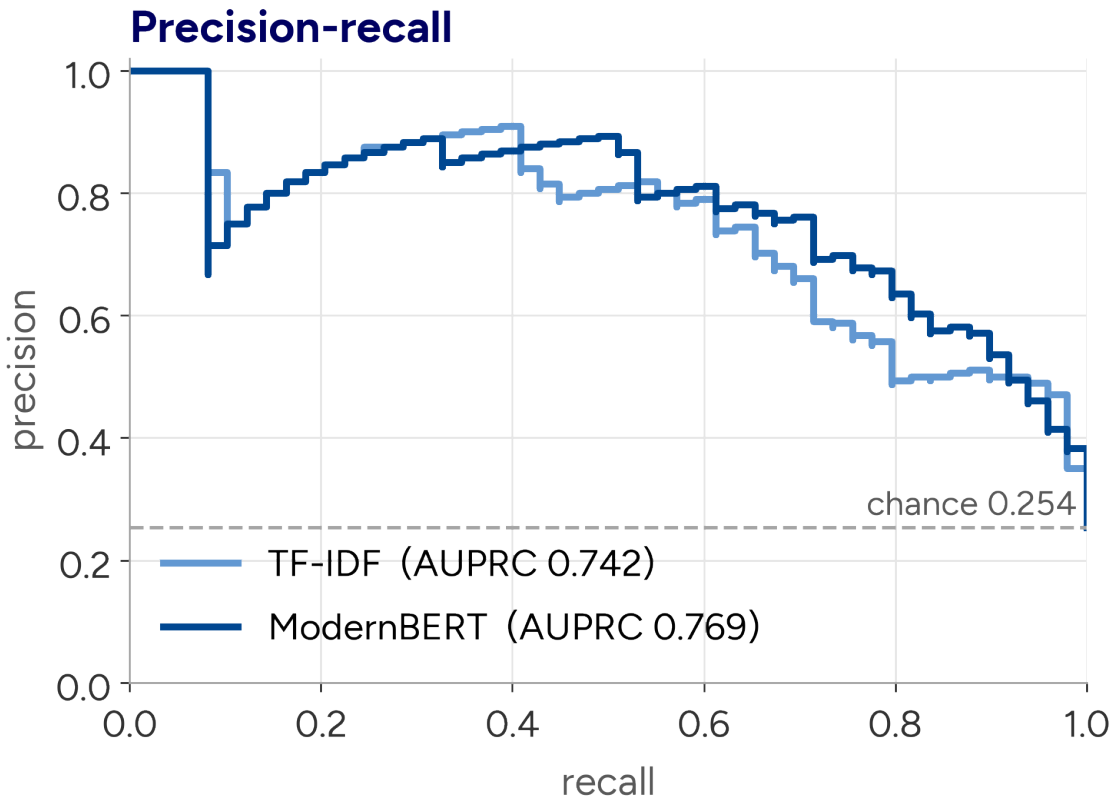
- naive_enough: the Naive answer was already enough to answer the question correctly.
- mix_required: the Naive answer was not enough, but the Mix answer provided more context that led to a correct answer.
- none_enough: neither answer was enough.

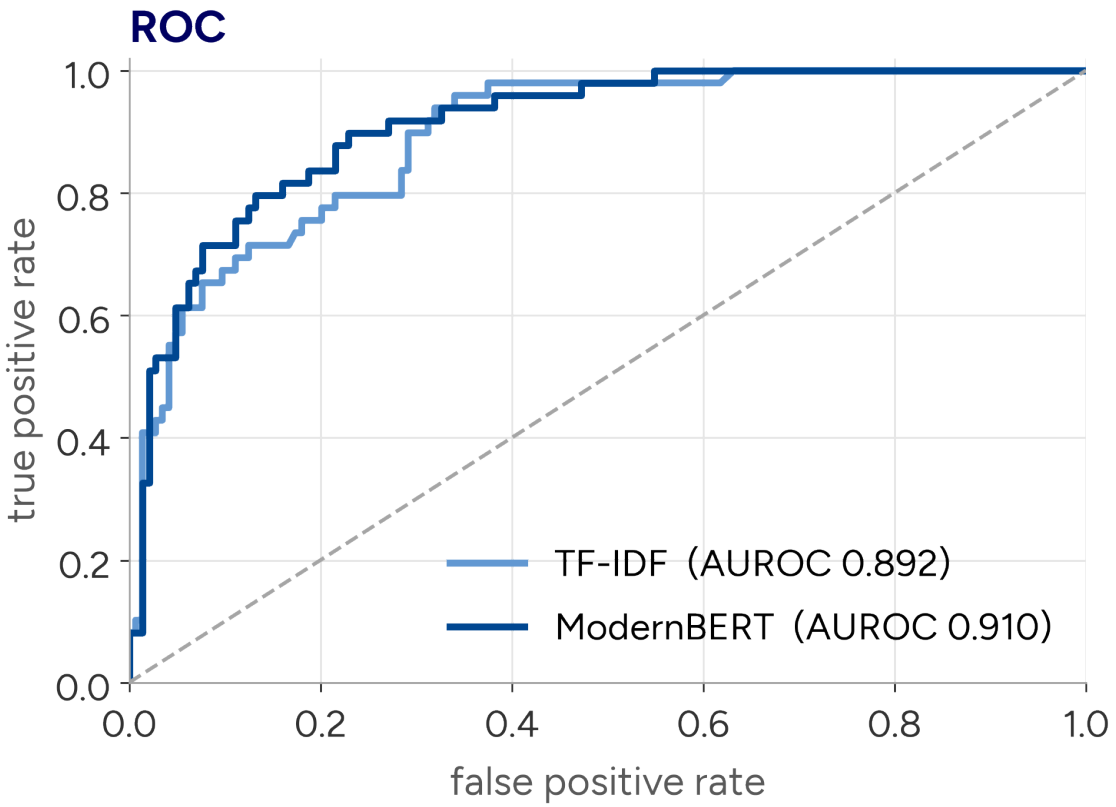
Key rules (9 in total)

- Use the ground-truth answer to judge correctness when it is provided. A longer answer is not better just because it is longer.
- If both Naive and Mix are correct, choose naive_enough.
- Treat explicit admissions of missing information and obvious error messages as not enough.

Model gemini-3-flash-preview, temperature 0, one fixed prompt for all 2,000 questions. Full system and user prompt verbatim in thesis Appendix B.

Backup: precision-recall and ROC curves on the test set





Backup: why the ranking differences stay undecided

Metric	Observed difference	Min. detectable difference (80% power, α = 0.05)	Test set needed
AUROC	+0.018	≈ 0.04	≈ 5× larger
Balanced accuracy	+0.018	≈ 0.08	≈ 21× larger
AUPRC	+0.027	≈ 0.11	≈ 17× larger
F1 (mix)	+0.068	≈ 0.13	≈ 4× larger

Derived from the paired-bootstrap standard errors on 193 test questions (49 Mix-required). A non-significant result here means undecided, not equal. Required size scales with (detectable / observed)².

Backup: even a perfect Naive-first cascade costs more time than query-only routing

Policy	Mean time per query	Est. tokens per query	Kept answers*	Wait for Mix-needing questions
ModernBERT router (query-only)	**14.1 s**	**6,666**	**91.7%**	**31 s (Mix only)**
Perfect cascade: Naive first, escalate exactly the Mix-required	19.5 s	7,992	100%	59 s (Naive, then Mix)
Always-Mix	24.4 s	17,857	100%	24 s

A cascade pays for Naive on every question, and its escalated questions pay for both modes. The perfect cascade is the cheapest possible full-recall cascade; a real one needs an imperfect failure detector, so it would cost more or keep fewer answers.

Estimate from the per-query times and per-mode token means on the 193 test questions; not part of the thesis. *Kept answers under the same assumption as slide 17. Mix-needing wait: mean over the questions each policy sends to Mix.

Backup: experimental configuration

Component	Setting
Generator and judge	gemini-3-flash-preview, temperature 0
Embeddings	gemini-embedding-001 (768-dim)
LightRAG	v1.4.10; chunks 1,200 tokens, overlap 100; Naive top_k = chunk_top_k = 5; Mix chunk_top_k = 5 (graph retrieval at LightRAG defaults)
TF-IDF router	uni- and bigrams, min df 2, 20k features; logistic regression, C = 1.0, balanced weights
ModernBERT router	ModernBERT-base, max 128 tokens, lr 2e-5, batch 16, up to 8 epochs, early stopping on validation F1
Seeds and thresholds	seeds 7, 13, 42; thresholds from validation F1 (TF-IDF 0.48, ensemble 0.44)
Code and data	github.com/Dantecool02/Thesis_ericsson, tag examiner-revision (labels, splits, outputs in results/)

Backup: use of generative AI tools

OpenAI Codex and Anthropic Claude Code were used in an advisory role, as discussion partners:

- discussing phrasing and implementation alternatives, and debugging
- preparing the revisions after supervisor and examiner feedback, and this presentation

The research question, design, experiments, analyses and conclusions are my own; every reported number was checked against the logs and output files.

Thesis Section 5.7, Use of generative AI tools. The Gemini model inside the pipeline is part of the method, not of this disclosure.